F(A, B, C, D) = Σm(2, 4, 5, 12, 14) + Σd(11, 13)

NEW FUNCTION

2. Finding Prime Implicants

Group	Decimal	Binary
0	2	0010 ✷
0	4	0100 ✓
1	5	0101 ✓
1	12	1100 ✓
2	14	1110 ✓
2	11	1011 ✷
2	13	1101 ✓

3. Finding Prime Implicants

Group	Decimal	Binary
0	12, 4	-100 ✓
0	4, 5	010- ✓

Group	Decimal	Binary
1	13, 5	-101 ✓
1	12, 14	11-0 ✷
1	12, 13	110- ✓

4. Finding Prime Implicants

Group	Decimal	Binary
0	12, 13, 4, 5	-10- ✷

The Prime Implicants Are

A'B'CD', AB'CD, ABD', BC'

6. Prime Implicants Chart

Minterms	12	14	2	4	5	Cost
A'B'CD'			X			8
ABD'	X	X				5
BC'	X			X	X	4

7. Finding Unique Minterms

Minterms	12	14	2	4	5	Cost
A'B'CD'			X			8
ABD'	X	X				5
BC'	X			X	X	4

The Essential Implicants Are

A'B'CD', ABD', BC'

Possible Function Minimizations

F = A'B'CD' + ABD' + BC'